

## Lecture 5 – Optimizers Practice Quiz

- Q1. In the local linear view of optimization, what does the gradient  $\nabla_{\theta} L(\theta)$  tell you about the loss around the current parameters?
- Q2. Why do we need an optimizer at all when training neural networks, instead of just randomly changing the weights?
- Q3. What is the basic update rule for vanilla gradient descent, and what does the learning rate control?
- Q4. Why is training time often described as “\$\$\$” in the context of choosing and tuning optimizers?
- Q5. Intuitively, why might sign SGD be more robust to very large or very small gradient magnitudes than standard gradient descent?
- Q6. In plain language, what does the RMS norm of a weight vector measure, and why is that natural in the context of optimization?
- Q7. Geometrically, what does the spectral norm of a weight matrix tell you about how that matrix acts on input vectors?
- Q8. From a geometric or physical perspective, what is Shampoo trying to fix about standard gradient descent?
- Q9. Explain the chain:  
Xavier initialization  $\rightarrow$  RMS norm  $\rightarrow$  induced matrix norm  $\rightarrow$  spectral norm  $\rightarrow$  Shampoo.  
Give a 3–5 sentence narrative of how each link in this chain motivates the next one.
- Q10. In the SVD-based view of Shampoo, the gradient  $\nabla W$  is written as  $U \Sigma V^T$  and the update is approximated as  $W \leftarrow W - \eta U \Sigma V^T$ . What does each part of this expression represent?
- Q11. What is the main problem  $\mu P$  (maximal update parametrization) tries to solve when we scale a model?
- Q12. Why might someone training a large language model prefer using Muon (a Shampoo-like optimizer with  $\mu P$ ) over standard optimizers?

## Answer Key

- A1. In the local linear view, the gradient  $\nabla_{\theta} L(\theta)$  gives the direction in parameter space in which the loss increases most rapidly.
- A2. Randomly changing the weights is extremely inefficient because the parameter space is huge and most changes do not lead to a decrease in loss.
- A3. The basic gradient descent update is  $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$ , where  $\eta$  is the learning rate. The learning rate controls how big each step is in the direction of negative gradient.
- A4. Training time is “\$\$\$” because running many optimization steps on large models requires a lot of computation.
- A5. Sign SGD only keeps the sign of each gradient coordinate and throws away the exact magnitude, so it is more robust to large or small gradients.
- A6. The RMS norm of a weight vector measures the “typical size” or average magnitude of its entries: roughly  $\sqrt{\text{average of squares}}$ .
- A7. Geometrically, the spectral norm of a matrix is the maximum factor by which the matrix can stretch a vector.
- A8. Shampoo is trying to correct for the fact that in standard gradient descent, the gradient is measured in the Frobenius norm, which is not aligned with the geometry of the parameter space.
- A9. Xavier initialization starts from the goal that each layer should keep activations at a reasonable scale, which leads to a specific initialization rule.
- A10. In the SVD  $\nabla W = U \Sigma V^T$ ,  $U$  and  $V$  encode the main directions of variation, while  $\Sigma$  encodes how much each direction contributes to the total update.
- A11.  $\mu P$  focuses on the problem that hyperparameters like the learning rate often “break” when we scale a model.
- A12. Someone training an LLM might prefer Muon because it can significantly reduce training time (often by a factor of 2 or more) compared to standard optimizers.